SOFTWARE REQUIREMENTS SPECIFICATION

AI Career Guidance and Skill Recommendation System

Project Title	AI Career Guidance and Skill Recommendation System
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1. Introduction

1.1 Purpose

This SRS defines the functional, non-functional, interface, data, and modeling requirements for an AI Career Guidance and Skill Recommendation System. The system helps students and learners make informed career decisions by analyzing education, interests, skills, experience, preferences, and assessment responses.

1.2 Scope

The system collects a learner profile and assessment responses, evaluates them through a recommendation engine, suggests suitable career paths, produces a suitability score, identifies skill gaps, recommends priority skills and learning resources, and tracks learning progress.

1.3 Objectives

- Provide personalized career recommendations.
- Explain why a career is recommended.
- Identify gaps between current and target-career skills.
- Recommend practical skills and learning resources.
- Track progress toward a selected career goal.

2. Overall Description

2.1 Intended Users

User	Main Responsibilities
Student / Learner	Creates profile, completes assessment, views career recommendations, checks skill gaps, follows learning recommendations, and tracks progress.
Career Counselor	Reviews learner information and recommendations to support mentoring and career planning.
Administrator	Manages users, career roles, required skills, learning resources, and system data.

2.2 Product Perspective

The system is a personalized decision-support platform. User information is converted into structured profile data, evaluated against career and skill knowledge, and returned as ranked career recommendations, suitability scores, skill gaps, and learning priorities.

2.3 High-Level System Flow

User Profile + Assessment → **AI Recommendation Engine** → **Career Ranking** → **Suitability Score** → **Skill-Gap Analysis** → **Skill Recommendations** → **Learning Resources** → **Progress Tracking**

2.4 Major Product Functions

Authentication, profile management, assessment, AI career recommendation, score generation, skill-gap analysis, skill recommendation, learning-resource recommendation, progress tracking, dashboard/reporting, and administrator data management.

3. Functional Requirements

ID	Requirement	Description
FR1	Registration & Login	The system shall allow users to register and securely authenticate.
FR2	Profile Management	The system shall allow users to enter and update education, interests, skills, experience and preferences.
FR3	Career Assessment	The system shall collect assessment responses related to interests, aptitude, strengths and career preferences.
FR4	Career Recommendation	The system shall analyze the profile and rank suitable career paths.
FR5	Suitability Score	The system shall generate a score or confidence indicator for each recommended career.
FR6	Skill-Gap Analysis	The system shall compare current skills with required career skills and identify missing or weak skills.
FR7	Skill Recommendation	The system shall recommend priority skills based on identified gaps and the selected career.
FR8	Learning Resources	The system shall recommend relevant courses, projects, certifications or practice resources.
FR9	Progress Tracking	The system shall record skill-learning progress and allow users to update completion status.
FR10	Dashboard & Reports	The system shall display recommendations, scores, gaps, resources and progress in a consolidated dashboard.
FR11	Career Data Management	The administrator shall be able to add and update career roles, required skills and career metadata.
FR12	Resource Management	The administrator shall be able to add and update learning resources and skill mappings.

4. Non-Functional Requirements

ID	Quality Attribute	Requirement
NFR1	Security	User credentials and profile data shall be protected from unauthorized access.
NFR2	Performance	Normal recommendation requests should return results within an acceptable interactive response time.
NFR3	Reliability	The system shall provide consistent results for the same input and handle failures gracefully.
NFR4	Usability	The interface shall be clear, responsive, and understandable for students.
NFR5	Scalability	The system shall support growth in users, career roles, skills and resources.
NFR6	Maintainability	Career/skill data and recommendation logic shall be modular and maintainable.
NFR7	Availability	The deployed application should be available whenever users need the service.
NFR8	Explainability	Recommendations should show relevant matching factors or skill gaps.
NFR9	Privacy	Only information necessary for career guidance shall be collected and access shall be role-controlled.

5. External Interface Requirements

5.1 User Interface

The web interface shall provide registration/login, profile setup, assessment, recommendation results, career details, skill-gap analysis, learning resources, and progress tracking. The layout shall be responsive.

5.2 Software Interface

The frontend shall communicate with backend services through APIs. The backend shall access the database and recommendation engine. The AI component shall accept structured user attributes and return ranked career and skill recommendations.

5.3 Communication Interface

Deployed communication shall use HTTPS. API requests and responses should use a structured format such as JSON.

6. System and Data Requirements

6.1 Hardware Requirements

Component	Minimum / Recommended
Processor	Intel Core i3 / equivalent or better
RAM	4 GB minimum; 8 GB recommended for local AI/ML development
Storage	At least 10 GB free space
Display	1366 × 768 or higher
Internet	Required for web deployment and external learning resources

6.2 Software Requirements

Component	Requirement
Operating System	Windows / Linux / macOS
Frontend	Web-based frontend application
Backend	Server-side application and REST API
Database	Database for users, skills, careers, assessments, recommendations and progress
AI/ML	Career and skill recommendation engine
Development	VS Code or equivalent; Git/GitHub
Browser	Chrome / Edge / Firefox or modern equivalent

6.3 Core Data Entities

Entity	Important Attributes
User	user_id, name, email, education, experience, preferences
Skill	skill_id, skill_name, category, proficiency_level
Career	career_id, title, description, required_skills, domain
Assessment	assessment_id, user_id, questions, responses, score
Recommendation	recommendation_id, user_id, career_id, score, reasons
Learning Resource	resource_id, title, skill_id, type, provider, link
Progress	progress_id, user_id, skill_id, status, completion_percentage

7. Data Flow Diagrams (DFD)

7.1 DFD Level 0 — Context Diagram

The complete application is represented as a single process. External entities provide user/profile information and receive recommendations, reports and guidance outputs.

DFD Level 0 — Context Diagram

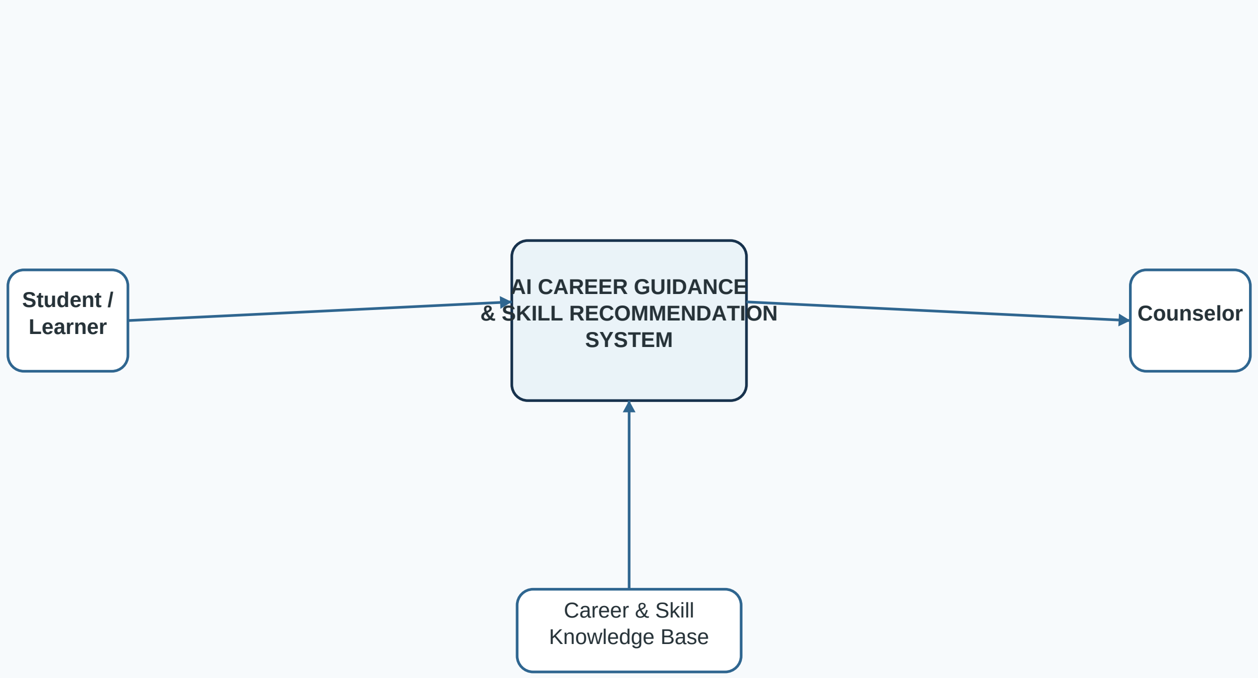

Figure 1. DFD Level 0 — Context Diagram

7.2 DFD Level 1 — Major Processes

Level 1 decomposes the system into profile/assessment, AI recommendation, skill-gap analysis, and progress/reporting processes. The main logical data stores are User Profiles and Career & Skills.

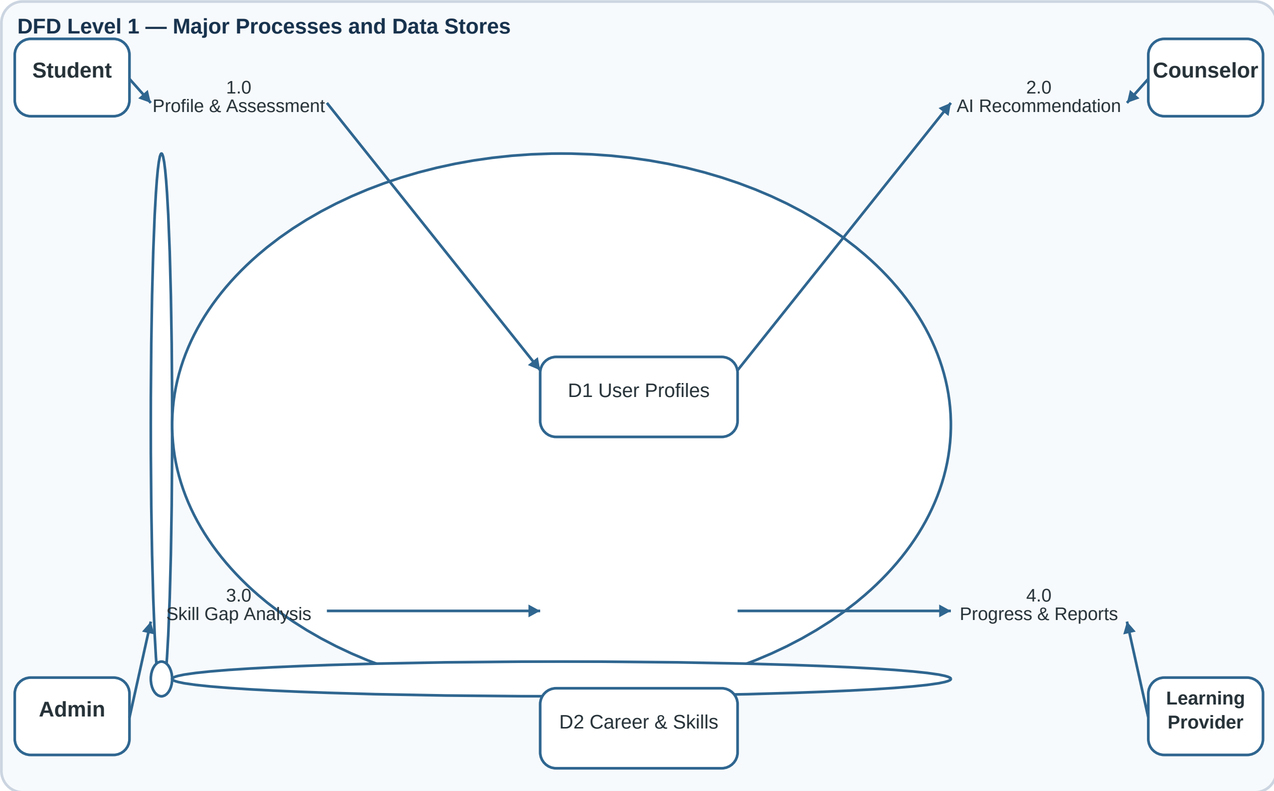

Figure 2. DFD Level 1 — Major Processes and Data Stores

8. Decision Table

The decision table defines how the system should interpret the combination of career match, skill gap, interest alignment, and profile completeness. It can be used as a rule layer alongside the AI recommendation model.

Rule	Career Match	Skill Gap	Interest Match	Profile Complete	Action
R1	High	Low	High	Yes	Recommend strongly; show skills and learning path.
R2	High	High	High	Yes	Recommend; prioritize missing/weak skills and create a roadmap.
R3	High	Low	Low	Yes	Show as possible option; explain interest mismatch and suggest reassessment.
R4	Low	—	—	Yes	Do not prioritize; show alternative higher-match careers.
R5	—	—	—	No	Request profile/assessment completion before final recommendation.

8.1 Decision Logic

A practical implementation may combine normalized factors such as skill match, interest match, education/experience fit, and assessment fit. A generic formulation is: **Overall Score = weighted skill match + interest match + education/experience fit + assessment fit**. Exact weights should be validated using project data.

8.2 Example Output

For a complete profile with high career and interest match but a large skill gap, the system should still recommend the career but emphasize the missing skills and present a learning roadmap.

9. Use Case Summary

Actor	Primary Use Cases
Student / Learner	Register/Login; Manage Profile; Take Assessment; Get Career Recommendations; View Score; View Skill Gap; Get Skill Recommendations; View Resources; Track Progress.
Career Counselor	Login; Review Profile; Review Recommendations; Interpret Skill Gaps; Support Career Planning.
Administrator	Login; Manage Users; Manage Career Roles; Manage Skills; Manage Learning Resources; View Reports.

10. Requirement Traceability Matrix

Requirement	Related Module / Model	Primary Actor
FR1–FR2	Authentication & Profile	Student
FR3	Assessment Module	Student
FR4–FR5	AI Recommendation Engine	Student / Counselor
FR6–FR7	Skill Gap & Skill Recommendation	Student / Counselor
FR8	Learning Resource Module	Student
FR9–FR10	Progress & Dashboard	Student / Counselor
FR11–FR12	Administration Module	Administrator
NFR1, NFR9	Security & Privacy	All
NFR2–NFR8	Application Quality	All

11. Constraints, Risks and Future Enhancements

11.1 Constraints

• Career and skill requirements change over time.
• Recommendation quality depends on the quality and coverage of the underlying data/model.
• External learning resources may change or become unavailable.
• Recommendations are guidance and are not guaranteed career outcomes.

11.2 Risks and Mitigations

Risk	Impact	Mitigation
Incomplete user data	Irrelevant recommendations	Validation, completeness checks and reassessment.
Outdated career/skill data	Incorrect roadmap	Administrator updates and periodic review.
Model bias	Unsuitable recommendations	Evaluate diverse profiles and expose matching factors.
Over-reliance on AI	Poor user decision-making	Clearly present results as guidance, not a final decision.

11.3 Future Enhancements

• Resume parsing and automatic skill extraction.
• Real-time job-market trend integration.
• Personalized multi-month career roadmaps.
• Mentor/counselor matching.
• Evidence-based skill assessment through projects or coding tests.
• Feedback-driven recommendation improvement.

12. Conclusion

The AI Career Guidance and Skill Recommendation System provides a structured approach to career planning. It combines profile and assessment information with career/skill knowledge to produce ranked career options, suitability scores, skill-gap analysis, prioritized skills, learning resources, and progress tracking. The DFDs and decision table provide clear software-engineering representations of data movement and decision logic.

End of Software Requirements Specification